INDUS UNIVERSITY

Monthly Progress Report - Software Group Project

(Report: 4)

Academic Year: 2026-27 Month: July

Project Title: House price prediction using AI/ML

Project Domain: AI/ML

Group Number: 50 Project Guide: Dhwani Mam

Student Details:

Enrolment No.	Student Name	Contribution	Signature
IU2341050103	Ravi Patel	➢ Deep Learning Dataset Preparation ➢ Artificial Neural Network (ANN) Architecture ➢ Hidden Layer Design ➢ Model Compilation	
IU2341050105	Kenil Patel	➢ ANN Performance Comparison with ML Models ➢ Final Model Selection ➢ ANN Model Saving (. karas) ➢ Deep Learning Result Analysis	
IU2341050114	Vishnu Jikadara	➢ ANN Training ➢ Training & Validation Graphs ➢ ANN Prediction ➢ ANN Model Evaluation	

Project Objective:

To develop and evaluate an Artificial Neural Network (ANN) for house price prediction, compare its performance with machine learning models, and select the best-performing model for deployment.

Monthly Work Completed

- Prepared the dataset for Deep Learning.
- Built an Artificial Neural Network (ANN) using TensorFlow and Keras.
- Designed input, hidden, and output layers.
- Compiled the ANN using Adam optimizer and Mean Squared Error (MSE) loss function.
- Trained the ANN model using the training dataset.
- Generated training and validation loss graphs.
- Generated training and validation MAE graphs.
- Predicted house prices using the trained ANN.
- Evaluated ANN using MAE, RMSE, and R^2 Score.
- Compared ANN performance with Linear Regression, Decision Tree, and Random Forest models.
- Selected the best-performing model based on evaluation metrics.
- Saved the trained ANN model (ann_model.keras).

Challenges Faced

- Understanding Deep Learning concepts and ANN architecture.
- Selecting appropriate hyperparameters (epochs, batch size, neurons).
- Monitoring overfitting using validation metrics.
- Comparing Deep Learning results with Machine Learning models.

Next Month Plan

- Develop Flask web application.
- Integrate the trained ANN model into the web application.
- Design user-friendly HTML interface.
- Test real-time house price prediction.
- Deploy the complete House Price Prediction System.

Guide Remarks

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Overall Progress : 60%

Student Signatures:

- Ravi Patel: _____
- Kenil Patel: _____
- Vishnu Jikadara: _____

Project Guide: _____